

Advance Docker Assignment

1) Use multi-stage builds

```
Dockerfile x docker-compose.yml README.md
1 # Build Stage
2 FROM node:20-alpine AS build
3
4 WORKDIR /app
5
6 COPY package*.json ./
7
8 RUN npm install
9
10 COPY . .
11
12 RUN npm run build
13
14 # Production Stage
15 FROM nginx:alpine
16
17 COPY --from=build /app/dist /usr/share/nginx/html
18
19 EXPOSE 80
20
21 CMD ["nginx", "-g", "daemon off;"]
```

2) Set up a custom Docker network and connect multiple containers to it

```
gaury@IN-1M96C54:/mnt/c/Project/docker assignment$ docker network create employee-network
4cd3ee89c22d5b0fa98b6451c59bb651c03202ca58ea5c7f00671433d8e0800b
gaury@IN-1M96C54:/mnt/c/Project/docker assignment$ docker network ls
NETWORK ID          NAME                                     DRIVER            SCOPE
b80c990bdcf2        bridge                                 bridge            local
15fcd3f2c912        commerce-platform-dev-local-env-setup_default   bridge            local
4cd3ee89c22d        employee-network                       bridge            local
48ec5480ef48        host                                  host              local
a465e407a16c        none                                  null              local
gaury@IN-1M96C54:/mnt/c/Project/docker assignment$
```

```

gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker network inspect employee-network
[
  {
    "Name": "employee-network",
    "Id": "c3a524c4a402d3a82f8252c180125d6fdc53f535f2e5a9dedd6f99706c9c4037",
    "Created": "2026-06-10T09:51:29.664252108Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
      "057d42bf9af0e2813cb7690b3db4954cd133852fb839b311249608a7e93381ed": {
        "Name": "react-app",
        "EndpointID": "887a5e1e75193f66033e24b883310bd6b749252165e434af0e62169a765cebf4",
        "MacAddress": "16:a5:5f:37:cb:9e",
        "IPv4Address": "172.20.0.4/16",
        "IPv6Address": ""
      },
      "85d169d0e94404f077b6831ae7afa372cb2770d681ec1dc86b823faceef945ab": {
        "Name": "docker-springbootapp",
        "EndpointID": "da2f4f478261dc53fba8f68256265f1c454e417c87e7eb021b9daff4ccde5940",
        "MacAddress": "a6:41:cd:e6:8b:75"
      }
    }
  }
]

```

3) Use Docker commands to inspect and manage the network.

```

gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker network ls
NETWORK ID          NAME                                     DRIVER  SCOPE
b80c990bdcf2        bridge                                 bridge  local
15fcd3f2c912        commerce-platform-dev-local-env-setup_default  bridge  local
c3a524c4a402        employee-network                      bridge  local
48ec5480ef48        host                                 host     local
a465e407a16c        none                                 null     local
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker network prune
WARNING! This will remove all custom networks not used by at least one container.
Are you sure you want to continue? [y/N] y
Deleted Networks:
commerce-platform-dev-local-env-setup_default

gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker network ls
NETWORK ID          NAME                                     DRIVER  SCOPE
b80c990bdcf2        bridge                                 bridge  local
c3a524c4a402        employee-network                      bridge  local
48ec5480ef48        host                                 host     local
a465e407a16c        none                                 null     local
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$

```

4) Create and manage Docker volumes to persist data for your containers. Use named volumes and bind mounts

```

gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker volume ls
DRIVER      VOLUME NAME
local       0f4b2715a3430226185ad6da22e903aa87ab52b0fb0aa1ebeb9e0fec9b106d3c
local       commerce-platform-dev-local-env-setup_mariadb_data
local       d93afa75e2ec4b315c1b7931b070bfafd677ef691b955a438db7bb47d546e73c
local       dockerassignment_mysql-data
local       dockertestcrud_mysql-data
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker volume inspect dockerassignment_mysql-data
[
  {
    "CreatedAt": "2026-06-10T07:23:22Z",
    "Driver": "local",
    "Labels": {
      "com.docker.compose.config-hash": "f4c855bfe9180ae6c444a7d2f638aafb341ecf2ee4a7bb1286c320e4aa3d1f40",
      "com.docker.compose.project": "dockerassignment",
      "com.docker.compose.version": "2.35.1",
      "com.docker.compose.volume": "mysql-data"
    },
    "Mountpoint": "/var/snap/docker/common/var-lib-docker/volumes/dockerassignment_mysql-data/_data",
    "Name": "dockerassignment_mysql-data",
    "Options": null,
    "Scope": "local"
  }
]
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker compose down
[+] Running 3/3
✔ Container react-app           Removed
✔ Container docker-springbootapp Removed
✔ Container mysql-container     Removed
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ mkdir backend-logs
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker compose up -d
[+] Running 3/3
✔ Container mysql-container     Started
✔ Container docker-springbootapp Started
✔ Container react-app           Started
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker exec -it docker-springbootapp sh
/app # echo "Docker Assignment" > /logs/test.txt
/app # exit
gauri@IN-1M96C54:/mnt/c/Project/docker assignment$ docker exec -it docker-springbootapp sh
/app # echo "testing volumn bind mount" > /logs/test.txt

```